Non-Extractive Capital

Digital Sovereignty Fund

Portfolio Simulation, Redemption Trigger, and Evergreen Cooperative Waterfall Framework

March 29, 2026

Working paper focused on portfolio-company simulation and evergreen cash-flow design

Contents

1. Purpose of This Note

This note isolates a new modelling question arising from Beata's request: how should the Digital Sovereignty Fund (DSF)simulate the development of a plausible portfolio company over time, from day 1 through early scale, in a way that is financially disciplined, operationally realistic, and consistent with the wider logic of the Fund.

The emphasis here is not yet on building a full stochastic engine. The immediate goal is more basic and more important: to define the **state variables**, **year-by-year operating profile**, and **success conditions** that a later spreadsheet model, system-dynamics model, or Monte Carlo simulation can use.

> This document should be read as a modular extension to the prior unified DSF model. It does not replace the financial, impact, or theological framework. Rather, it adds an intermediate layer: the simulated operating life of the portfolio company itself.

2. Framing Assumption

The portfolio company being modelled here is not a conventional venture-backed startup optimised for valuation inflation and exit. It is instead a digital-sovereignty-oriented company, often emerging from an open-source, infrastructure, or grant-adjacent context, which must become commercially viable without becoming extractive.

Accordingly, the simulation should not ask only whether a company can grow. It should ask whether it can grow **in the right way**. In practical terms, this means:

- early emphasis on technical credibility and real product usefulness,
- a controlled rather than performative burn rate,
- increasing capacity for procurement and institutional adoption,
- disciplined hiring rather than growth for growth's sake,
- movement toward break-even and retained resilience rather than dependence on perpetual capital injections.

> The right comparison class is not the hyper-growth Silicon Valley company. The right comparison class is the technically strong but commercially under-structured European open technology company that needs help converting usefulness into durable adoption.

3. What the Simulation Must Produce

At minimum, the simulation should generate the following outputs for each company profile:

1. capital required by year,
2. team size and team composition by year,
3. annual operating cost and monthly burn,
4. revenue ramp and timing of first meaningful customer income,
5. time to break-even,
6. probability of survival to each year,
7. cumulative company-level redemption potential under the Fund's capped-return logic,
8. adoption indicators relevant to digital sovereignty impact.

These outputs then feed forward into the higher-level portfolio model already developed elsewhere.

4. Core Modelling Logic

4.1. A Company as a State Vector

Let each company at time t be represented by a state vector

$$X_t = (N_t, W_t, O_t, R_t, C_t, Q_t, P_t, A_t, B_t),$$

where:

- N_t is team size,
- W_t is average annual salary cost per full-time equivalent,
- O_t is non-payroll operating expense,
- R_t is annual recurring or contracted revenue,
- C_t is available cash,
- Q_t is product maturity or implementation readiness,
- P_t is procurement access or sales capacity,

- A_t is adoption traction,
- B_t is burn rate.

The simulation does not need all variables to be stochastic on day 1. Some can initially be deterministic policy assumptions. The key point is to make the company legible as a dynamic system rather than merely as a valuation placeholder.

4.2. Operating Equations

The baseline accounting relations can remain simple.

$$\text{Payroll}_t = N_t \cdot W_t, \tag{1}$$
$$\text{Cost}_t = \text{Payroll}_t + O_t, \tag{2}$$
$$B_t = \frac{\text{Cost}_t - R_t}{12}, \tag{3}$$
$$C_{t+1} = C_t + I_t + R_t - \text{Cost}_t, \tag{4}$$

where I_t is capital injected by the Fund during period t.

A first-pass sustainability condition is then:

$$R_t \geq \text{Cost}_t.$$

This is economically simple, but it is strategically very important: it makes viability rather than valuation the key threshold.

4.3. Interpretation of EBITDA

If an EBITDA-style measure is included, it should be used as a **sustainability indicator**, not as a proxy for speculative enterprise value. The basic form is:

$$\text{EBITDA}_t = R_t - \text{Cost}_t,$$

with the understanding that, in this context, a move from deeply negative to mildly positive EBITDA is not merely an accounting milestone. It is evidence that the company is becoming operationally self-supporting.

> In the wider DSF logic, the moral significance of such a threshold is that the enterprise begins to stand more fully on the strength of its work and service rather than on the continuing subordination of its future to external capital claims.

5. Closing the Loop - From Company Cash Flow to Company Redemption and Cooperative Waterfall

Beata's request adds an essential next step. It is not enough to model company growth and survival in the abstract. The model must now close the financial loop by specifying:

1. when a portfolio company becomes capable of **redemption**,
2. how much of that cash flow returns to the Fund,
3. how the Fund recycles that money back into new deployments,
4. under what condition the Fund begins distributing to members, and
5. what happens once a member has fully received the capped return.

This requires three aligned layers of formulas: a **company-level redemption formula**, a **Fund-level recycling formula**, and a **member-level capped distribution formula**. The key design problem is not merely mathematical. It is temporal. Redeem too early and the company is weakened. Recycle too aggressively and member patience is strained. Distribute too early and the evergreen compounding logic is broken.

5.1. Company-Level Financial Chain

The first missing piece is a concrete path from revenue to free cash flow. For company i in year t, let:

$$\begin{aligned}
\mathrm{Revenue}_{i,t} &= R_{i,t}, && (5)\\
\mathrm{CashOpex}_{i,t} &= \mathrm{Payroll}_{i,t} + O_{i,t}, && (6)\\
\mathrm{EBITDA}_{i,t} &= R_{i,t} - \mathrm{CashOpex}_{i,t}, && (7)\\
\mathrm{EBIT}_{i,t} &= \mathrm{EBITDA}_{i,t} - D_{i,t}, && (8)\\
\mathrm{Tax}_{i,t} &= \tau_i \cdot \max\big(0, \mathrm{EBIT}_{i,t}\big), && (9)\\
\mathrm{FCF}_{i,t} &= \mathrm{EBITDA}_{i,t} - \mathrm{Tax}_{i,t} - \mathrm{Capex}_{i,t} - \Delta\mathrm{NWC}_{i,t}. && (10)
\end{aligned}$$

Here $D_{i,t}$ is depreciation and $\Delta\mathrm{NWC}_{i,t}$ is the change in net working capital. This gives the Fund a cash-flow measure that is much more useful than revenue alone. Revenue may rise while the company still has no safe capacity to redeem. EBITDA may turn positive while growth still absorbs cash. Free cash flow is therefore the right bridge variable between operating success and conditional equity return.

5.2. Operating Layer, Resilience Layer, and Conditional Allocation Layer

Beata's proposed logic is strongest when written explicitly as a three-layer waterfall inside the company itself.

(a) Operating layer.

Revenue is first absorbed by the ordinary life of the company:

$$\mathrm{FCF}_{i,t} = \mathrm{Revenue}_{i,t} - \mathrm{CashOpex}_{i,t} - \mathrm{Tax}_{i,t} - \mathrm{Capex}_{i,t} - \Delta\mathrm{NWC}_{i,t}.$$

Nothing is redeemable until this layer has already been satisfied.

(b) Resilience and solvency layer.

Let the minimum resilience reserve be

$$L^{\star}_{i,t} = \max\left(\overline{L}_i,\ \rho_i \frac{\mathrm{CashOpex}_{i,t}}{12},\ S_{i,t}\right).$$

Let $C^{\mathrm{pre}}_{i,t}$ be opening cash before the period's allocation decisions. Then the reserve gap is

$$\mathrm{ResGap}_{i,t} = \max\left(0, L^{\star}_{i,t} - (C^{\mathrm{pre}}_{i,t} + \mathrm{FCF}_{i,t})\right).$$

The company must first refill this gap before any redemption becomes thinkable. Here $S_{i,t}$ can stand for any explicit solvency, covenant, or board-imposed minimum cash requirement.

(c) Conditional allocation layer.

Only the remaining cash is available for conditional allocation:

$$\mathrm{DistributableCash}_{i,t} = \max\left(0, \mathrm{FCF}_{i,t} - \mathrm{ResGap}_{i,t}\right).$$

This amount is then split between internal reinvestment and investor redemption:

$$\mathrm{Reinvest}_{i,t} = \gamma_{i,t}\,\mathrm{DistributableCash}_{i,t}, \tag{11}$$

$$\mathrm{RedBase}_{i,t} = (1-\gamma_{i,t})\,\mathrm{DistributableCash}_{i,t}, \tag{12}$$

where $\gamma_{i,t} \in [0,1]$ is the company-level reinvestment share.

> This is the key shift in language and substance. The company is not servicing debt. It is conditionally allocating surplus cash between resilience, further growth, and the redemption of capped equity rights.

5.3. Redemption Trigger Event

A redemption event should not be triggered by revenue alone. A company can have meaningful revenue and still be too fragile to support Fund claims. A more prudent trigger is:

$$T_{i,t} = \mathbf{1}\Big(\mathrm{EBITDA}_{i,t} > 0,\ \mathrm{DistributableCash}_{i,t} > 0,\ \mathrm{CumRed}_{i,t-1} < \Omega_{i,t}\Big),$$

where:

- $\mathrm{CumRed}_{i,t-1}$ is cumulative redemption already made to the Fund,
- $\Omega_{i,t}$ is the maximum cumulative redemption obligation for company i.

A useful first specification for the company redemption cap is:

$$\Omega_{i,t} = \kappa_i \sum_{\tau \le t} I_{i,\tau},$$

where κ_i is the company-level redemption multiple and $I_{i,\tau}$ is DSF capital deployed into company i up to time t.

> It is important to separate the *company-level* cap κ_i from the *member-level* cap $r = 3$. The company instrument determines what the company may redeem back to the Fund. The cooperative cap determines what the Fund may ultimately distribute to its members. These do not need to be numerically identical, although they must be economically compatible.

5.4. Redemption Amount Formula

Once the trigger is active, the redemption amount should be linked to *distributable cash* rather than fixed amortisation. A simple first-pass rule is:

$$\mathrm{Red}_{i,t} = T_{i,t} \cdot \min\Big(\mathrm{RedBase}_{i,t},\ \Omega_{i,t} - \mathrm{CumRed}_{i,t-1}\Big), \tag{13}$$

$$\mathrm{CumRed}_{i,t} = \mathrm{CumRed}_{i,t-1} + \mathrm{Red}_{i,t}. \tag{14}$$

This formula is attractive for the DSF because it aligns with real company lifecycles:

- if the company is weak, redemption is zero,
- if the company is improving but still cash-tight, redemption is small,
- if the company is truly generating surplus cash, redemption accelerates naturally,

- the cap prevents the Fund claim from becoming permanently extractive.

A useful implementation refinement is to make the reinvestment share time-sensitive:

$$\gamma_{i,t} = \begin{cases} \gamma_i^{\text{early}}, & t \leq 3, \\ \gamma_i^{\text{late}}, & t \geq 4, \end{cases} \qquad \text{with } \gamma_i^{\text{early}} > \gamma_i^{\text{late}}.$$

This reflects the idea that the company should retain more internally during the fragile early phase and only later release a larger share toward redemption.

5.5. What Happens Once the Company-Level Cap Is Fully Redeemed

Once $\text{CumRed}_{i,t} = \Omega_{i,t}$, the investor's economic rights in respect of those particular shares cease. The relevant shares are then redeemed, cancelled, or converted into non-economic rights in accordance with the company's articles. No further company-level cash is payable in respect of those units unless the Fund subscribes for a *new* investment round, creating a new capital contribution and therefore a new capped economic claim.

> In this structure, redemption begins not when the company looks exciting, but when it can afford reciprocity without undermining resilience. The same resilience-first logic can then be lifted upward into the Fund and cooperative waterfall.

5.6. Illustrative Effect of the Layered Company Waterfall

Using the deterministic company example developed later in this note, the layered logic changes the interpretation of Years 3 and 4. The company does not simply redeem because EBITDA is positive. It first fills resilience needs, then retains a reinvestment share, and only then redeems.

Year	FCF	Reserve gap	Distrib. cash	Reinvest share	Redemp.
1	€-345k	€0k	€0k	–	€0k
2	€-332k	€0k	€0k	–	€0k
3	€163k	€40k	€123k	60% = €74k	€49k
4	€532k	€0k	€532k	60% = €319k	€213k

The point is not the exact numbers. The point is the mechanic. Once the resilience layer is made explicit, the company becomes easier to track, easier to simulate, and easier to govern with discipline.

6. Three Portfolio Archetypes

A single "average company" is useful for exposition but too crude for serious simulation. The portfolio should eventually be modelled using at least three archetypes.

6.1. Archetype I - Infrastructure Core

This archetype includes foundational software, protocols, security tools, and other digital public goods that may be strategically vital while remaining commercially awkward.

Typical features:

- small, highly technical team,
- relatively slow revenue ramp,
- high strategic or ecosystem value,
- longer path to commercial self-sufficiency.

6.2. Archetype II - Service and Integration Layer

This archetype includes support, implementation, managed service, compliance, and integration companies built around useful open technology.

Typical features:

- somewhat faster route to paying customers,
- stronger visibility into unit economics,
- medium growth but comparatively high reliability.

6.3. Archetype III - Scalable Platform or Product Layer

This archetype includes productised offerings with more cross-border scaling potential.

Typical features:

- stronger upside,
- greater hiring and go-to-market complexity,

- more volatile outcomes,
- higher need for disciplined anti-extractive governance as scale accelerates.

> For the first simulation draft, it is reasonable to begin with one stylised baseline company. But the model should be built so that archetype-specific parameter sets can be swapped in immediately once calibration begins.

7. Baseline Startup and Scale-up Profile

The table below proposes a pragmatic first-pass operating profile for a stylised DSF portfolio company. The values are not presented as historical truth. They are modelling inputs intended to be refined through real company examples, including candidate NLnet-related companies.

Stage	Team and operating profile	Indicative annual cost	Indicative annual revenue
Day 1	2 - 4 highly technical founders; limited commercial capacity; often grant-adjacent or undercapitalised; no structured sales function	€180k - €300k	€0 - €120k
Year 1	4 - 6 people; first product and operations support; initial business development capacity; first procurement conversations	€350k - €500k	€50k - €200k
Year 2	6 - 10 people; clearer implementation capacity; first repeatable customer profile; more structured delivery	€600k - €1.0m	€200k - €600k
Year 3	10 - 18 people; recognisable market presence; delivery, support, and sales functions now distinct	€1.1m - €1.8m	€700k - €1.5m
Year 4	15 - 25 people; sustainable operating discipline; procurement pipeline broad enough to support resilience	€1.6m - €2.5m	€1.5m - €3.0m+

7.1. Suggested Team Composition by Stage

The simulation should not track headcount alone. It should track composition, since this is what determines whether growth is coherent.

Stage	Technical	Product	Ops	Sales / Partnerships	Support / Delivery
Day 1	2 - 4	0 - 1	0	0	0
Year 1	3 - 4	1	0 - 1	0 - 1	0
Year 2	4 - 6	1	1	1 - 2	0 - 1
Year 3	6 - 10	1 - 2	1	1 - 2	1 - 3
Year 4	8 - 14	2	1 - 2	2 - 3	2 - 4

7.2. Indicative Cost Decomposition

To make the simulation operational, annual cost should be decomposed into at least:

$$\mathrm{Cost}_t = N_t W_t + O_t^{\mathrm{infra}} + O_t^{\mathrm{legal}} + O_t^{\mathrm{travel}} + O_t^{\mathrm{sales}} + O_t^{\mathrm{misc}}. \tag{15}$$

This matters because not all operating expenses scale at the same speed. Infrastructure and legal costs may be front-loaded, while sales and travel may accelerate later when procurement and deployment activity increase.

8. Entry-Level Parameters for a First Spreadsheet Model

A useful initial spreadsheet or simulation dashboard should expose the following parameters explicitly, rather than hiding them in hard-coded formulas.

8.1. Founding Conditions

- initial team size N_0,
- initial cash on hand C_0,
- average salary W_0,
- initial annual operating expense O_0,
- initial annual revenue R_0,
- first DSF ticket size I_0.

8.2. Growth and Execution Parameters

- hiring lag by role,

- annual salary inflation,
- time to first meaningful customer,
- annual revenue growth rate,
- revenue per customer or contract,
- customer retention rate,
- probability of procurement access,
- probability of delayed sales conversion.

8.3. Governance and Fund Parameters

- maximum total capital deployed,
- follow-on trigger conditions,
- redemption cap multiple,
- redemption start threshold,
- minimum cash runway requirement.

9. A Minimal Deterministic Scenario Engine

Before introducing randomness, the first useful simulation is deterministic. This allows the team to inspect the causal logic directly.

9.1. Step-by-Step Annual Update

For each year t, the spreadsheet or code should perform the following sequence:

1. update team composition and resulting payroll,
2. update operating expenses,
3. update product maturity and go-to-market readiness,
4. update customer count and resulting revenue,
5. compute burn and cash remaining,
6. test whether additional capital must be deployed,

7. test whether the company remains alive,
8. test whether the company has reached break-even.

9.2. Simple Revenue Construction

The revenue side can initially be expressed as:

$$R_t = K_t \cdot Y_t,$$

where K_t is the number of paying customers and Y_t is average annual revenue per customer. This is intentionally plain. It is much better to begin with a transparent formula and complicate only when real company evidence demands it.

10. A Monte Carlo-Ready Extension

Once the deterministic skeleton works, selected parameters can be turned into random variables.

10.1. Candidate Random Variables

A first Monte Carlo version could sample from distributions for:

- time to first revenue,
- annual revenue growth,
- conversion from pipeline to signed contracts,
- hiring delays,
- salary escalation,
- unexpected operating cost shocks,
- retention and churn.

10.2. Illustrative Simulation Structure

Let company i in simulation run s evolve according to sampled parameters $\theta_i^{(s)}$. Then the engine computes:

$$X_{t+1}^{(i,s)} = F\left(X_t^{(i,s)}, \theta_i^{(s)}, I_t^{(i,s)}\right),$$

for $t = 0, 1, 2, 3, 4$. Across many simulation runs, the model can then report empirical distributions for survival, capital need, break-even timing, and capped company-level redemption.

10.3. Why This Matters for the Fund

At portfolio level, Monte Carlo modelling would make it possible to estimate:

- expected number of viable companies after 4 years,
- distribution of required follow-on capital,
- average time to self-sustaining operation,
- redemption distribution under the Fund's cap,
- sensitivity of outcomes to better procurement access or better customer conversion.

11. Portfolio Alignment - The Three Formulas Beata Is Asking For

The Fund now needs three formulas that are distinct but aligned.

11.1. Formula I - Company to Fund

At portfolio-company level, the Fund receives *redemption proceeds* from operating cash flow. Aggregating across all companies gives annual Fund inflow:

$$\Pi_t = \sum_{i=1}^{n_t} \mathrm{Red}_{i,t}.$$

This is the first formula Beata is asking for: the return of cash from companies into the Fund pot.

11.2. Formula II - Cooperative Waterfall and Evergreen Recycling

Let the cooperative's gross proceeds in period t be

$$\mathrm{GrossProceeds}_t = \Pi_t + \mathrm{LiquidationProceeds}_t + \mathrm{OtherRealisedProceeds}_t.$$

The first layer of the cooperative waterfall is then:

$$\mathrm{NetProceeds}_t = \max\big(0, \mathrm{GrossProceeds}_t - \mathrm{CoopOpex}_t - \mathrm{CoopTax}_t - \mathrm{CoopLiabilities}_t\big), \tag{16}$$

$$\mathrm{AvailAfterReserve}_t = \max\big(0, \mathrm{NetProceeds}_t - \mathrm{ReserveAlloc}_t\big). \tag{17}$$

This corresponds directly to Beata's cooperative logic: first operating costs, taxes, and liabilities; second the Board's reserve allocations; only then a split between reinvestment and member distributions.

Let E_t denote the evergreen pot at the start of period t, and let $E_t^\star$ denote the minimum target level required to preserve the compounding engine. Define the policy reinvestment share

$$\eta_t = \begin{cases} \eta^{\text{early}}, & t \le 3, \\ \eta^{\text{late}}, & t \ge 4, \end{cases} \qquad \text{with } \eta^{\text{early}} > \eta^{\text{late}}.$$

To ensure the evergreen pot is never underfilled merely because the policy share was set too low, define actual reinvestment as

$$\mathrm{ReinvestFund}_t = \min\Big(\mathrm{AvailAfterReserve}_t,\ \max\big(\eta_t\, \mathrm{AvailAfterReserve}_t,\ E_t^\star - E_t\big)\Big).$$

Then the amount available for member distribution is

$$\mathrm{DistPool}_t = \mathrm{AvailAfterReserve}_t - \mathrm{ReinvestFund}_t.$$

Finally the evergreen stock evolves as

$$E_{t+1} = E_t + \mathrm{ReinvestFund}_t - \mathrm{NewDeploy}_t.$$

This is the mechanism Beata was reaching for. The cooperative may adopt an *intended* reinvestment split such as "x% in the first three years and z% thereafter," but the actual formula should still override that split whenever more retention is needed to protect the evergreen engine.

11.3. Formula III - Fund to Members Under the Capped Return Rule

For member m and subscription vintage v, let:

- $K_{m,v}$ be contributed capital,
- $u_{m,v}$ be participation units,
- $\mathrm{CumDist}_{m,v,t-1}$ be cumulative distributions already received on that vintage,

- $r_{m,v}$ be the applicable cap multiple, with the current baseline being $r_{m,v} = 3$.

The remaining economic headroom of that vintage is

$$H^{\text{cap}}_{m,v,t} = r_{m,v} K_{m,v} - \text{CumDist}_{m,v,t-1}.$$

Let $\mathcal{V}_t$ be the set of *live* subscription vintages for which headroom remains positive. Then the provisional pro rata payout is

$$\widetilde{\text{Dist}}_{m,v,t} = \frac{u_{m,v}}{\sum_{(a,b)\in\mathcal{V}_t} u_{a,b}} \cdot \text{DistPool}_t,$$

and the actual payout is

$$\text{Dist}_{m,v,t} = \min\bigl(\widetilde{\text{Dist}}_{m,v,t},\ H^{\text{cap}}_{m,v,t}\bigr).$$

Any residual amount created when one vintage hits the cap is redistributed pro rata among the remaining live vintages, and any amount that still cannot be distributed is retained by the cooperative and folded back into reinvestment.

11.4. What Happens Once the Capped Return Has Been Fully Paid Out

Once a member's subscription vintage has received distributions equal to the applicable capped return, the economic rights attached to that vintage cease. At that point the member has the three options Beata described:

1. subscribe for *new* participation units, thereby creating a new capital contribution, a new economic vintage, and potentially even a new cap multiple,
2. remain in the cooperative in a *non-economic* capacity, preserving whatever non-economic participation, governance, community, or advisory rights the articles may allow,
3. withdraw from the cooperative in accordance with the articles.

This gives the structure a genuinely open-ended character. Members are not forced into an exit culture. They may remain as allies, advisors, or community participants even after their economic rights on a given capital vintage have been exhausted.

11.5. Rolling Investors in an Open-Ended Cooperative

The cleanest way to model rolling investors is therefore by *vintages*. Each new subscription (m, v) carries its own:

- contribution amount $K_{m,v}$,
- participation units $u_{m,v}$,
- cap multiple $r_{m,v}$,
- cumulative distribution clock $\mathrm{CumDist}_{m,v,t}$,
- status variable $s_{m,v,t} \in \{\text{economic}, \text{non-economic}, \text{exited}\}$.

The cooperative is thus open-ended without being vague. New money can come in later, even on new terms, while old money can remain socially and institutionally connected after its economic rights have ended.

> A very plausible first operating principle is therefore: *company-level redemptions flow first into the Fund pot; the cooperative first covers its own costs and reserves; then it retains enough to protect the evergreen engine; only the remainder becomes distributable to live economic vintages, pro rata, until each vintage reaches its cap.*

12. Illustrative Deterministic Company Example

The following table shows a stylised five-stage company profile under the formulas above. It is not a historical dataset. It is a worked example designed to show how company-level redemption begins only after revenue growth has translated into real free cash flow.

Assumptions for this illustration:

- two DSF investments: €400k in Year 1 and €350k in Year 2,
- company-level redemption multiple $\kappa = 2.0$ so the maximum company obligation is €1.5m,
- reinvestment share $\gamma = 60\%$ in Years 3 and 4 after the resilience layer,
- simplified tax rate $\tau = 25\%$ applied to positive EBIT.

Stage	Team	Revenue	Cash Opex	EBITDA	Tax	Capex	ΔWC	DSF In	Redemp.
Day 1	3	€40k	€286k	€-246k	€0k	€20k	€0k	€0k	€0k
Year 1	5	€180k	€480k	€-300k	€0k	€35k	€10k	€400k	€0k
Year 2	8	€520k	€772k	€-252k	€0k	€60k	€20k	€350k	€0k
Year 3	12	€1,550k	€1,172k	€378k	€85k	€90k	€40k	€0k	€49k
Year 4	18	€2,700k	€1,744k	€956k	€224k	€120k	€80k	€0k	€213k

The example makes the sequencing visible. The company shows meaningful revenue in Years 1 and 2 but still has no redemption capacity. Only once EBITDA is positive and cash generation survives tax, capex, working-capital needs, reserve rebuilding, and the deliberate reinvestment share does redemption to the Fund begin. By the end of Year 4, cumulative redemption is still modest relative to the original DSF deployment, which is exactly why the Fund-level recycling and distribution timing rules matter so much.

13. Why EBITDA Matters - But Is Not Enough

Beata's question about EBITDA is exactly right. EBITDA should be included because it captures whether the company has crossed from operational deficit toward operating surplus. However, EBITDA alone is still too optimistic for redemption design. A company may report positive EBITDA while remaining cash-hungry because:

- tax becomes payable,
- implementation and infrastructure require capex,
- working capital expands with growth,
- management still needs a resilience buffer.

Therefore the model should use the chain

$$\text{Revenue} \to \text{EBITDA} \to \text{EBIT} \to \text{Tax} \to \text{FCF} \to \text{Redemption}.$$

This is the company-level formula Beata is asking to make concrete.

14. Waterfall Interpretation for the Fund

Putting all three layers together, the intended waterfall becomes:

1. portfolio companies generate revenue and eventually free cash flow,
2. a governed share of distributable free cash flow is redeemed into the Fund pot,
3. the Fund first restores and protects its evergreen reserve,
4. only surplus above that reserve becomes distributable,
5. distributions are then made pro rata to members, subject to the 3x cap.

This gives a coherent answer to the strategic balancing problem:

- **survival and company success** come first,
- **Fund recycling and compounding** come second,
- **investor distributions** begin only after evergreen break-even is credibly secured.

> This is precisely the balance the DSF should be able to simulate: not extraction delayed by rhetoric, but a genuinely non-extractive circulation of capital in which company resilience, Fund perpetuity, and investor patience are mathematically coordinated.

15. Minimum Viable Structure for a First Launch

Beata's next request pushes the model one level closer to implementation. Instead of discussing a diversified future Fund in the abstract, we can define a **minimum viable legal and financial stack** that is sufficient to begin operating:

- one Coöperatie as the investing and distribution vehicle,
- one Stichting to hold the steward layer,
- one golden-share holder or equivalent mission-protection actor,
- one or two initial investor-members,
- one initial portfolio company,
- and, for now, one assumed loan from NPV of €300k.

This is enough to start modelling real cash movement without pretending that the full portfolio already exists.

15.1. Entity Stack

Entity	Role in the minimum viable structure
Coöperatie	Receives member capital, receives the NPV loan, makes the initial portfolio-company investment, receives redemptions and other proceeds, and applies the cooperative waterfall.
Stichting	Holds steward rights or analogous mission-locking rights, helping ensure that governance continuity is not reducible to investor economics.
Golden-share holder	Holds a veto-style or reserved right over specified mission-critical matters. This could be a trusted external steward institution or equivalent protective actor.
Initial members	For simulation purposes, one or two early investors are enough. A simple starting case is two tickets of €100k each.
Portfolio company	One initial company is enough to make the model operational, even though it is obviously under-diversified at portfolio level.
NPV loan	A provisional €300k loan that supplements the initial member capital and increases the first deployable pool, while also introducing an additional Fund-level liability that must be reflected in the waterfall.

15.2. Initial Capital Stack

Let the first-launch capital stack be:

$$K_0^{\text{members}} = \sum_{m \in \mathcal{M}_0} K_m, \qquad L_0^{\text{NPV}} = \text{initial NPV loan principal},$$

with the concrete first-pass assumption

$$K_0^{\text{members}} = 2 \times €100\text{k} = €200\text{k}, \qquad L_0^{\text{NPV}} = €300\text{k}.$$

Then initial gross cash at the cooperative level is

$$F_0^{\text{gross}} = K_0^{\text{members}} + L_0^{\text{NPV}} = €500\text{k}.$$

This gross amount is *not* the same as immediately deployable investment capital. Before the first company is funded, the structure must still absorb formation and prudential costs.

Let:

- $C_0^{\text{coop-setup}}$ be Coöperatie setup and launch cost,
- $C_0^{\text{stichting}}$ be Stichting setup cost,
- C_0^{golden} be the cost of implementing the golden-share or equivalent governance layer,

- $E_0^{\star}$ be the initial evergreen reserve the Board wishes to preserve from day 1.

Then net deployable cash before the first company commitment is

$$F_0^{\text{deploy}} = F_0^{\text{gross}} - C_0^{\text{coop-setup}} - C_0^{\text{stichting}} - C_0^{\text{golden}} - E_0^{\star}.$$

> The main practical lesson is simple: even in the smallest viable launch, the first ticket should not be modelled as if the full €500k is available for company operations. The fund stack itself consumes capital and also needs an initial buffer.

15.3. Steward-Ownership Condition for Portfolio Companies

Beata's request also adds a crucial structural condition: the portfolio companies themselves should be steward-owned, or become steward-owned, before the Fund takes them on. This means the model needs a **conversion layer** rather than pretending the company is already in its target legal form.

Let company i have a steward-ownership status indicator

$$Z_{i,t}^{\text{SO}} \in \{0,1\},$$

where $Z_{i,t}^{\text{SO}} = 1$ means the company is already SO-compliant at closing or becomes compliant as part of the transaction package. Then a simple investment admissibility rule is:

$$I_{i,t}^{\text{gross}} > 0 \implies Z_{i,t}^{\text{SO}} = 1 \text{ at or immediately after closing.}$$

This does not mean the company must arrive in perfect legal form before the Fund ever engages it. It means the model should treat the transition as part of the funded process. Let:

- C_i^{SO} be legal and structural transition cost into steward ownership,
- C_i^{inc} be incorporation or restructuring cost associated with the company-level vehicle,
- I_i^{gross} be the headline investment amount approved by the Fund.

Then the net operating capital actually reaching the company is:

$$I_i^{\text{net}} = I_i^{\text{gross}} - C_i^{\text{SO}} - C_i^{\text{inc}}.$$

This formula matters because otherwise the simulation will overstate the company's true starting runway. A €400k ticket is not economically a €400k operating ticket if €25k–€75k must first be spent on conversion, legal implementation, governance design, or new entity setup.

For modelling discipline, the clean default is therefore: book the full approved ticket at the Fund level, but treat the steward-ownership transition cost as a deduction from the company's usable operating capital unless the Fund explicitly finances it in a separate line item.

15.4. Single-Company Launch Formulas

For a minimum viable launch with exactly one initial portfolio company $i = 1$, define:

$$G_1 = \text{gross approved company allocation}, \qquad I_1^{\text{net}} = G_1 - C_1^{\text{SO}} - C_1^{\text{inc}}.$$

The launch feasibility condition is then:

$$G_1 \leq F_0^{\text{deploy}}.$$

If the first company also requires working-capital protection at closing, one may impose a stronger condition:

$$G_1 + B_0^{\text{fund}} \leq F_0^{\text{deploy}},$$

where B_0^{fund} is a Board-determined retained buffer at the cooperative level.

This yields a small but coherent first-launch stack:

1. raise €200k member capital,
2. add €300k NPV loan,
3. pay formation and governance setup costs,
4. hold back a minimum reserve,
5. allocate one gross ticket to one company,
6. deduct SO transition and incorporation costs from the gross ticket to get net company runway,
7. start observing revenue, resilience, reinvestment, redemption, and Fund-level recycling.

15.5. Illustrative Launch Budget

The following is a purely illustrative first-pass budget. Its role is not to settle final legal or commercial choices, but to make visible how much capital is genuinely left for operations once the minimal structure and company transition costs are acknowledged.

Illustrative source or use	**Amount**
Member capital: Investor A	€100k
Member capital: Investor B	€100k
NPV loan	€300k
Total initial gross cash	**€500k**
Coöperatie setup, documentation, and launch	€15k
Stichting setup and steward layer implementation	€10k
Golden-share layer / mission-protection implementation	€5k
Initial cooperative reserve retained	€50k
Net deployable before first company	**€420k**
Gross first company ticket G_1	€420k
Less steward-ownership transition cost C_1^{SO}	€25k
Less incorporation / restructuring cost C_1^{inc}	€15k
Net operating capital at company level I_1^{net}	**€380k**

The effect is immediate and important. Even in this tiny launch structure, a nominal €420k first commitment becomes only €380k of real operating capital once the company is moved into the required steward-owned configuration.

15.6. How the NPV Loan Changes the Mechanics

The NPV loan boosts the size of the initial deployable pool, but it also introduces a liability that should sit *ahead of member distributions*. In modelling terms, this means the cooperative waterfall should not only cover generic costs and taxes; it should also include scheduled debt service or principal restoration associated with the NPV loan.

Let:

$$\mathrm{DS}_t^{\mathrm{NPV}} = \mathrm{Int}_t^{\mathrm{NPV}} + \mathrm{Prin}_t^{\mathrm{NPV}}$$

be the period-t debt-service amount associated with the NPV loan. Then the cooperative net-proceeds formula becomes:

$$\mathrm{NetProceeds}_t = \mathrm{GrossProceeds}_t - O_t^{\mathrm{coop}} - \mathrm{Tax}_t^{\mathrm{coop}} - \mathrm{Liab}_t^{\mathrm{other}} - \mathrm{DS}_t^{\mathrm{NPV}}.$$

This changes the mechanics in four ways:

- the first company can be larger or can have more runway than member capital alone would allow,
- the cooperative now has an explicit financing liability that must be serviced before member distributions,
- the evergreen compounding rule becomes more delicate because part of early proceeds may need to replenish Fund strength *and* service the loan,
- member payout should be delayed not only until the evergreen engine is credible, but also until the Fund is not being weakened by premature leakage while an external loan still sits on the balance sheet.

A conservative simulation rule is therefore:

$$\mathrm{DistPool}_t > 0 \implies E_t \geq E_t^\star \text{ and } \mathrm{DS}_t^{\mathrm{NPV}} \text{ has been fully provided for.}$$

15.7. Why This Minimal Structure Is Useful for Simulation

This stripped-down structure is intentionally small, but analytically it is powerful. It lets the model represent all of the following at once:

- governance complexity at Fund level,
- steward-protection mechanics at both Fund and company level,
- a mixed capital stack of member equity-like capital plus a loan,
- the deduction of company transition costs from headline ticket size,
- and the timing tension between company resilience, Fund recycling, loan servicing, and eventual member distributions.

For simulation purposes, that is already enough structure to start learning. One can later add more members, more companies, more vintages, and more complex reserve rules without changing the underlying logic.

> The first simulation worth building is therefore not a ten-company portfolio. It is this minimum viable stack with one company, one real transition cost, one reserve target, and one external loan. If the logic does not work there, it will not work at scale.

16. Link Back to the Unified DSF Model

This company-level simulation layer should eventually sit between the existing financial and impact equations.

Fund design
capital policy
repayment cap → portfolio company
simulation engine
team, revenue, burn → adoption, sovereignty
and repay-
ment outcomes

In other words, the earlier DSF framework tells us what the Fund is trying to do. This note tells us how a company inside that framework should be represented so that the Fund's assumptions become testable.

17. Recommended First Build Order

A practical sequencing would be:

1. build one baseline deterministic company,
2. translate it into a 5-year spreadsheet,
3. calibrate it using a handful of real NLnet-relevant examples,
4. split the model into the three archetypes,
5. only then add Monte Carlo randomness.

This ordering reduces confusion. It makes the assumptions visible before statistical sophistication is layered on top.

18. Conclusion

The crucial modelling shift is this: the Digital Sovereignty Fund (DSF)should not begin by simulating valuation. It should begin by simulating **operating life**. The company must be represented as a living system with people, salaries, hiring choices, procurement delays, customer conversion, burn discipline, tax, capex, and cash retention. Only then can the portfolio model say something credible about capital needs, sustainability, sovereign adoption, Fund recycling, and capped investor return.

This revised note now closes a much larger part of the loop Beata identified. It provides:

- a concrete company-level path from revenue to EBITDA to free cash flow,
- an explicit operating layer, resilience layer, reinvestment amount, and redemption amount at company level,
- a prudential redemption trigger and redemption amount formula,
- a cooperative-level evergreen recycling rule,
- an open-ended member distribution rule by subscription vintage,
- post-cap options for members to reinvest, remain as non-economic allies, or leave,
- a minimum viable launch structure with one Co"operatie, one Stichting, one golden-share layer, one initial portfolio company, and one assumed €300k NPV loan,
- and an explicit deduction of steward-ownership transition cost from the company's usable capital at closing.

The next phase of work is therefore well defined: calibrate these formulas using a small number of plausible company archetypes, then run deterministic and Monte Carlo scenarios to test whether the timing and magnitude of company redemption, cooperative recycling, and member distributions remain aligned under realistic survival and growth assumptions.

A. Compact Variable Reference

Variable	Meaning
N_t	Team size at time t
W_t	Average annual salary cost per full-time equivalent
O_t	Non-payroll operating expense
R_t	Annual revenue
C_t	Cash available
I_t	Capital injected by the Fund during period t
B_t	Monthly burn rate
Q_t	Product maturity or implementation readiness
P_t	Procurement access or sales capacity
A_t	Adoption traction
K_t	Number of paying customers
Y_t	Average annual revenue per customer
$D_{i,t}$	Depreciation for company i in period t
τ_i	Effective tax rate used for company i
$\text{Capex}_{i,t}$	Capital expenditure for company i in period t
$\Delta\text{NWC}_{i,t}$	Change in net working capital for company i in period t
$\text{FCF}_{i,t}$	Free cash flow for company i in period t
$L^{\star}_{i,t}$	Minimum resilience reserve required at company level
$\text{ResGap}_{i,t}$	Cash amount needed to refill the company resilience reserve
$\text{DistributableCash}_{i,t}$	Company cash available after the resilience layer is satisfied
$\gamma_{i,t}$	Share of distributable company cash retained for reinvestment
$\text{RedBase}_{i,t}$	Company cash notionally available for redemption before cap check
$T_{i,t}$	Redemption trigger indicator for company i in period t
κ_i	Company-level redemption multiple
$\Omega_{i,t}$	Maximum cumulative redemption obligation of company i
$\text{Red}_{i,t}$	Actual company-level redemption paid to the Fund in period t
$\text{CumRed}_{i,t}$	Cumulative redemption already paid by company i
Π_t	Aggregate redemption inflow from all companies into the Fund
GrossProceeds_t	All cooperative proceeds realised in period t
NetProceeds_t	Cooperative proceeds remaining after its own costs, taxes, and liabilities
ReserveAlloc_t	Amount allocated by the Board to reserves at cooperative level
$\text{AvailAfterReserve}_t$	Proceeds available for the reinvestment-versus-distribution split
E_t	Evergreen Fund pot at the start of period t
$E^{\star}_t$	Minimum evergreen target that should be preserved or rebuilt
η_t	Target cooperative reinvestment share in period t
ReinvestFund_t	Actual cooperative amount retained for evergreen reinvestment
DistPool_t	Amount available for member distribution after the cooperative waterfall
$K_{m,v}$	Contributed capital for member m in subscription vintage v
$u_{m,v}$	Participation units for member m in vintage v
$r_{m,v}$	Cap multiple applicable to vintage (m, v)
$\text{CumDist}_{m,v,t}$	Cumulative distributions paid to vintage (m, v)
$H^{\text{cap}}_{m,v,t}$	Remaining distribution headroom before a vintage reaches its cap
$s_{m,v,t}$	Status of a member vintage: economic, non-economic, or exited
K^{members}_0	Aggregate initial member capital committed to the cooperative at launch
L^{NPV}_0	Initial NPV loan principal used in the first-launch capital stack
F^{gross}_0	Gross cooperative cash available at launch before setup costs and reserves
$C^{\text{coop-setup}}_0$	Cooperative formation, documentation, and launch cost
$C^{\text{stichting}}_0$	Stichting setup and steward-layer implementation cost
C^{golden}_0	Cost of implementing the golden-share or equivalent mission-protection layer
F^{deploy}_0	Net deployable cash available before the first portfolio-company commitment
$Z^{\text{SO}}_{i,t}$	Indicator that company i is steward-ownership compliant at time t
C^{SO}_i	Company-level steward-ownership transition cost
C^{inc}_i	Company incorporation or restructuring cost at transaction stage
I^{gross}_i	Headline approved investment amount for company i
I^{net}_i	Net usable operating capital reaching company i after transition costs
G_1	Gross approved first-company ticket in the minimum viable launch case
B^{fund}_0	Additional cooperative buffer retained at launch beyond the evergreen reserve
DS^{NPV}_t	Debt service associated with the NPV loan in period t